

Nate Woodward

LC T P Spring symposium: Theoretical Physics and AI

Datasets & Problems

1

3

Operational Difficulty

Easy

Medium

Hard

Domain Topic

ADVANCED UNDERGRADUATE

- Foundations and Scalar Fields
- Path Integrals and Quantization
- Symmetries and Scattering

GRADUATE

- Fermions and Spinor Structure
- Gauge Fields and Symmetry
- Renormalization and Corrections

ADVANCED GRADUATE

- Renormalization Group and Scaling
- Non-Abelian Gauge Theories
- Finite Temperature/Vacuum Effects

POST GRADUATE

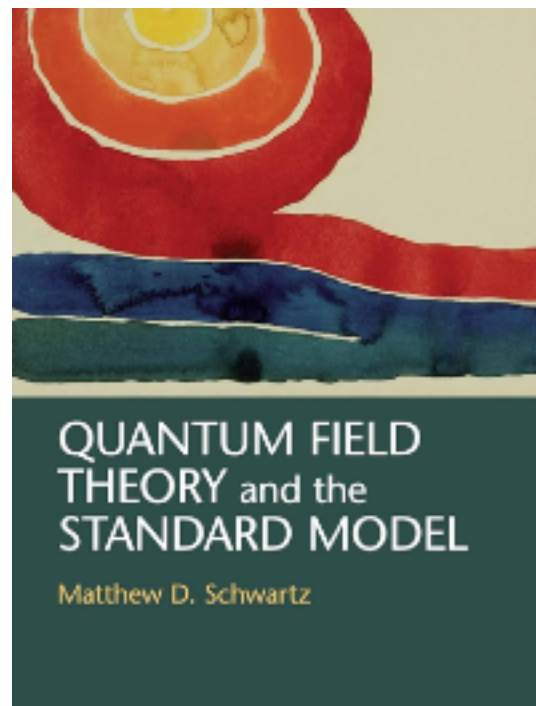
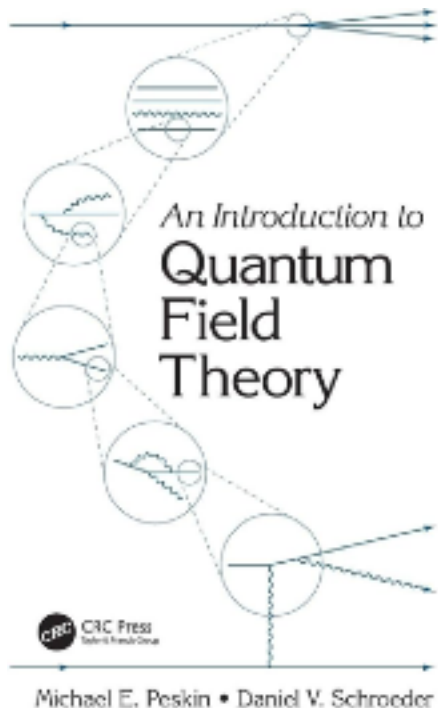
- Amplitude and On-Shell Methods
- Loop and Regularization Techniques
- Effective Field Theories

Synthetic

Human-Adapted

~~arXiv~~

QFT Pedagogy



...



Datasets & Problem Seeds

Synthetic

Domain Topic

ADVANCED UNDERGRADUATE	GRADUATE
• Foundations and Scalar Fields • Path Integrals and Quantization • Symmetries and Scattering	• Fermions and Spinor Structure • Gauge Fields and Symmetry • Renormalization and Corrections
ADVANCED GRADUATE	POST GRADUATE
• Renormalization Group and Scaling • Non-Abelian Gauge Theories • Finite Temperature/Vacuum Effects	• Amplitude and On-Shell Methods • Loop and Regularization Techniques • Effective Field Theories

Operational Difficulty

Easy Medium Hard

Human-Adapted



QFT Pedagogy



Our Data Pipeline

